

Youwei Xiao

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Education

Bachelor of Science degree in Computer Science

School of Electronics Engineering and Computer Science, Peking University

September 2018 - July 2022

Beijing, China

Ph.D. Candidate in EDA

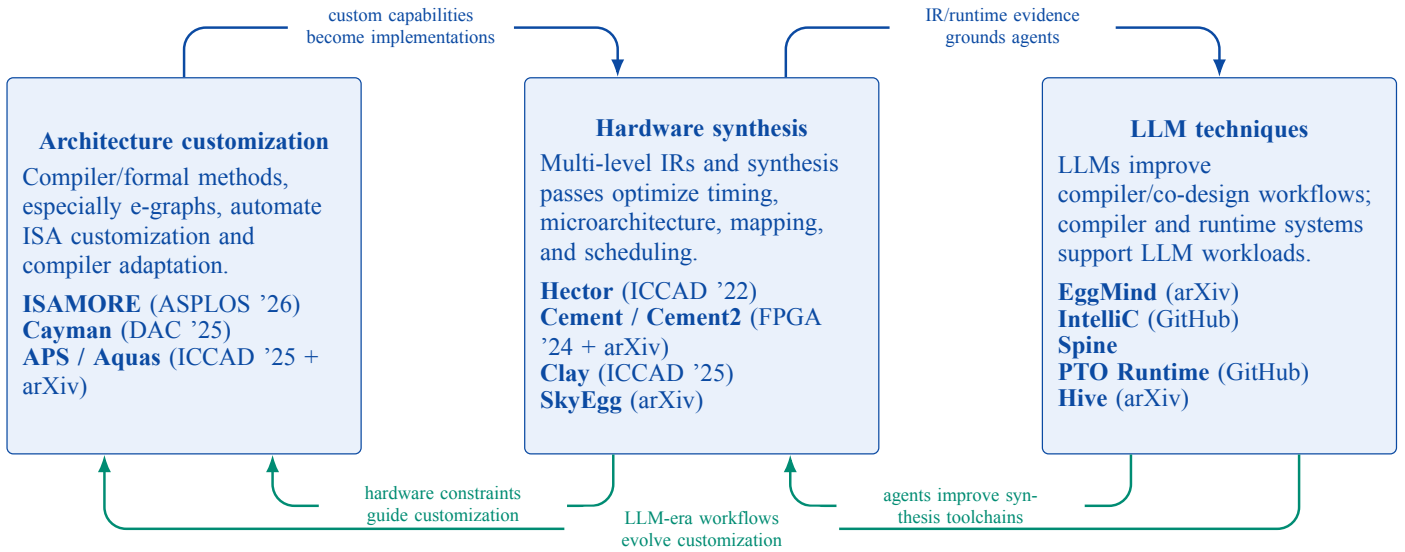
School of Integrated Circuits, Peking University

September 2022 - now

Beijing, China

Research

My research builds compiler-centered hardware-software co-design systems for domain-specific computing. The central problem is how real software demands can be translated into reusable architectural capabilities, realized as high-quality hardware, and exposed back to software through compiler interfaces that remain analyzable, verifiable, and reusable.



Architecture customization. I explored compiler techniques and formal methods, especially e-graphs, to automate ISA customization and compiler adaptation. *ISAMORE* (DOI) finds reusable custom instructions by anti-unifying equivalent program fragments. *Cayman* (DOI) discovers accelerator opportunities from whole applications while optimizing control flow and data access. *APS* (DOI, GitHub) and *Aquas* (preprint) connect customized hardware capabilities to retargetable compiler support.

Hardware synthesis. I created intermediate representations at multiple abstraction levels and proposed synthesis passes that optimize timing, microarchitecture, mapping, and scheduling. *Hector* (DOI, GitHub) provides a multi-level MLIR substrate for hardware synthesis. *Cement* (DOI) and *Cement2* (preprint) make cycle timing and temporal transactions first-class FPGA design abstractions. *Clay* (DOI) synthesizes microarchitecture-aware ASIP instructions, while *SkyEgg* (preprint) uses e-graphs for joint implementation selection and scheduling.

LLM techniques. I explored both LLMs for compilers and compiler/system techniques for LLMs. *EggMind* (preprint) uses LLMs to synthesize reusable equality-saturation strategies. *IntelliC* (GitHub) studies inspectable compiler representations for human and agent collaboration. *Spine* studies verification-bounded agentic co-design, using typed semantic boundaries and executable checks to connect design intent, architecture descriptions, hardware/compiler artifacts, and validation feedback in a traceable closed loop. *PTO Runtime* (GitHub) executes compiled task graphs for serving on Ascend chips and LingQu SuperPods. *Hive* (preprint) extends the systems view to multi-agent inference infrastructure.

The closed loop is that architecture customization exposes reusable hardware capabilities, hardware synthesis turns them into efficient implementations, and LLM-era techniques advance the earlier compiler and co-design toolchains while providing evolving capabilities that make the loop practical and self-improving.

Publications

- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26). doi:10.1145/3779212.3790162. **[Best Paper Award, 5/1048]**
- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240817. [Invited Paper]
- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240669. (*Equal contribution)
- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC '25). doi:10.1109/DAC63849.2025.11132875.
- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE '25). paper.
- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25). doi:10.1145/3706628.3708861.
- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD '24). doi:10.1145/3676536.3676830.
- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '24). doi:10.1145/3626202.3637561.
- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD '22). doi:10.1145/3508352.3549370.

Preprints

- Chenyun Yin*, **Youwei Xiao***, Yuze Luo, Yuyang Zou, and Yun Liang. LLM-Guided Strategy Synthesis for Scalable Equality Saturation. arXiv:2604.17364. (*Equal contribution)
- Zizhang Luo, Yuhao Luo, **Youwei Xiao**, Yansong Xu, Runlin Guo, and Yun Liang. Hive: A Multi-Agent Infrastructure for Algorithm- and Task-Level Scaling. arXiv:2604.17353.
- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. arXiv:2511.22267.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Honors and Awards

学术之芯 Awarded to 8 students for academic excellence	2026 School of IC, Peking University
博士研究生校长奖学金 Awarded to 10 students in School of IC, Peking University	2026-2027 Peking University
ASPLOS 2026 Best Paper Award Finding Reusable Instructions via E-Graph Anti-Unification	March 2026 5/1048 submissions
Second Place in EDATHON 2020 Programming Competition on Electronic Design Automation	August 2020 IEEE CEDA Hong Kong Chapter
Shenzhen Stock Exchange Scholarship Awarded to 17 students from EECS, Peking University	2019-2020
Merit Student Awarded to top 10% students for comprehensive excellence	2018-2019 & 2019-2020
Yang Fuqing & Wang Yangyuan Academician Scholarship Awarded to only 5 students at EECS, Peking University, for academic excellence	2018-2019
Second Prize in NOI 2017 National Olympiad in Informatics	July 2017 China Computer Federation
Second Prize in APIO 2017 Asia Pacific Informatics Olympiad (China District)	May 2017 China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework ASP-DAC 2026	January 2026 Hong Kong, China
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design ISED 2025	May 2025 Hong Kong, China
Agile Hardware Specialization (AHS) ASPLOS 2025	March 2025 Rotterdam, Netherlands
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design DATE 2025	March 2025 Lyon, France
AHS: An EDA toolbox for Agile Chip Front-end Design ASP-DAC 2025	January 2025 Tokyo, Japan